Robotics Data Hackathon — Problem Statement

The Problem

Robotics and humanoid AI are rapidly growing, but there is a major bottleneck:

Robots cannot learn effectively from simulations alone.
They need real-world human activity data — especially videos of people performing physical tasks.

This type of data is called egocentric (first-person) video data, and currently:

- **Large-scale robotics datasets are mostly created outside India**
- **There is no structured pipeline in India for collecting and preparing such data**
- **Huge demand exists for high-quality, labeled human interaction data**

This hackathon challenges you to build that pipeline from scratch

Your Objective

Build a complete robotics data pipeline using egocentric video data.

This includes:

1. **Collecting or using sample video data**
2. **Segmenting actions (start/end of tasks)**
3. **Annotating objects + hand interactions**
4. **Generating natural language descriptions**
5. **Structuring a dataset usable for AI models**

Sample Data

You can either:

- **Collect your own egocentric videos**
  OR
- **Use the provided sample dataset**

Sample Videos (Google Drive):

[Hackathon (Raw) - Google Drive]

[Factory recordings - Google Drive]

Key Concept: Egocentric Data

- Third-person video: Camera watches a person
- Egocentric video: Camera sees what the person sees

Why important?

- Robots learn from their own viewpoint
- Egocentric data = more natural learning for robots

What You Can Build

You may choose to focus on:

Option A — Data Pipeline Creation

- Collect + process real-world videos
- Build a structured dataset

Option B — AI / Analysis

- Build tools for annotation / QA
- Train action recognition models
- Improve dataset quality

Expected Pipeline

A strong solution should include:

1. Video Collection / Input
2. Preprocessing (cleaning, trimming)
3. Action Segmentation
4. Object + Hand Annotation
5. Text Description Generation
6. Dataset Structuring (JSON / format)
7. Quality Validation

What Matters Most

Judging will focus on:

- Data Quality — clean, consistent annotations
- Pipeline Completeness — end-to-end system
- Problem Understanding — clarity of approach
- Demo / Output — working result

Why This Matters

- Robotics is a multi-billion dollar industry
- AI models need massive real-world data
- This is a foundational problem few teams are solving

You are not just building a project —
you are building infrastructure for future robots

Goal

Create something that:

- A robot learning model can directly use
- Demonstrates real-world applicability
- Can scale beyond the hackathon

Build data. Train robots. Shape the future.